Beyond Demographics: Fine-tuning Large Language Models to Predict Individuals' Subjective Text Perceptions

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Abstract

People naturally vary in their annotations for subjective questions and some of this variation is thought to be due to the person's sociodemographic characteristics. LLMs have also been used to label data, but recent work has shown that models perform poorly when prompted with sociodemographic attributes, suggesting limited inherent sociodemographic knowledge. Here, we ask whether LLMs can be trained to be accurate sociodemographic models of annotator variation. Using a curated dataset of five tasks with standardised sociodemographics, we show that models do improve in sociodemographic prompting when trained *but* that this performance gain is largely due to models learning annotator-specific behaviour rather than sociodemographic patterns. Across all tasks, our results suggest that models learn little meaningful connection between sociodemographics and annotation, raising doubts about the current use of LLMs for simulating sociodemographic variation and behaviour.

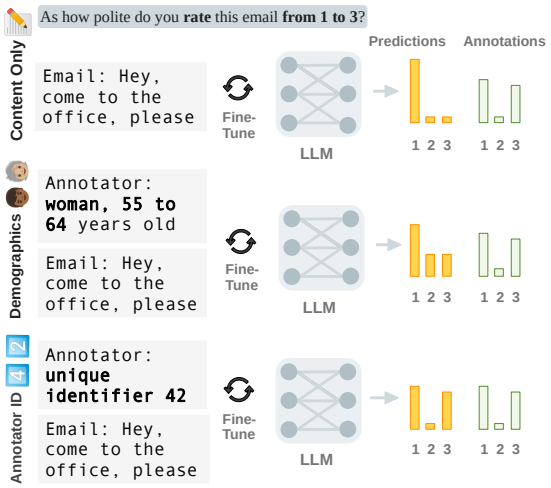


Figure 1: We test if LLMs can be *trained* to predict individuals' subjective text perceptions based on their demographic background, finding that models rather model specific individuals than learn from demographics.

1 Introduction

Most Natural Language Processing (NLP) models require labelled data to learn. Yet, the humans labelling that data may not agree what is the correct label. These annotator disagreements stem from multiple causes, such as genuine mistakes, adversarial behaviour, or personal preferences (Röttger et al., 2022; Sandri et al., 2023). This variance in labelling behaviour has long been recognised and multiple models have been developed to distinguish some types of disagreements, particularly those due to unreliable annotators (Hovy et al., 2013; Passonneau and Carpenter, 2014). Recent work has focused on modelling the regularity in label variation due to individual (e.g., Deng et al., 2023) and group-based preferences (e.g., Davani et al., 2024). For example, members of one social group may regularly rate a piece of text as more or less offensive than others. When such labelling behaviour is regular, a large language model (LLM) could be prompted to take on sociodemographic characteristics to generate how a person with such characteristics would answer (Beck et al., 2024).

Such approaches of sociodemographic prompting require that an LLM can effectively take the perspective of a person or group. When LLMs are used to synthetically label data (Fröhling et al., 2024) or as evaluators (Dong et al., 2024), this approach potentially provides a scalable way to include meaningful variation by annotators, particularly those for less common sociodemographic identities (Simmons and Hare, 2023). However, multiple works have raised issues with the accuracy of this approach in zero-shot settings (e.g., Hu and Collier, 2024; Sun et al., 2025). While the root cause of this low zero-shot performance is likely multifaceted, given the potential benefits of LLMs as annotator models, **we test whether LLMs can be trained as sociodemographic models of annotators**, which was not assessed before.

Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 2092–2111
July 27 - August 1, 2025 ©2025 Association for Computational Linguistics

To effectively model individual annotators with LLMs, we introduce a new approach that combines sociodemographic prompting with annotator modelling. Instead of fixing annotator identity (ID) and attributes as part of a specialised architecture, we incorporate this information by adapting input formats from sociodemographic prompts. Using this formatted input, we fine-tune decoder-only LLMs with prediction heads as used in reward models for LLM alignment (e.g., Liu et al., 2024). For training, we curated the DEMO dataset by unifying five existing datasets for subjective classification tasks (intimacy, offensiveness, politeness, safety, sentiment) that include annotator IDs and sociodemographic attributes (age, gender, race, and education).

Our study answers the following four research questions. **RQ1**: Can LLMs learn to model *given* annotators better based on sociodemographics or their identity (ID)? LLMs improve over baselines when incorporating sociodemographics, but we find that LLMs are much more accurate at modelling specific annotators' behaviours. **RQ2**: Can LLMs generalise to *new* annotators? No, we find that neither sociodemographic attributes nor IDs improve performance over a text-only baseline, suggesting LLMs do not learn generalisable patterns. **RQ3**: If LLMs can use sociodemographic attributes to better model given annotators (RQ1) but do not generalise (RQ2), what information do LLMs learn when improving from sociodemographics? We find that sociodemographic-tuned models primarily improve for annotators with unique attributes, where attributes effectively act as an ID, suggesting models primarily improve at modelling specific individuals. **RQ4**: Does modelling annotator identity improve how models predict label distributions when annotators disagree? Beyond improvements in predicting annotations, we show that models using annotator identity better reflect cases of disagreement between annotators than models using sociodemographics.

Together, these results highlight that models do not learn substantial patterns between annotator sociodemographics and annotation behaviour. Based on this finding, we caution against using current LLMs to replicate text perceptions of individuals from specific social groups, in particular if no examples of their actual behaviours are available. All data and code related to our experiments are available at https://github.com/morlikowski/beyond-demographics.

2 Related Work

Within research on the role of annotator characteristics in annotation, we connect work in sociodemographic prompting with annotator modelling.

Annotator Characteristics in Annotation. Training NLP and AI models relies on human annotations to adjust their parameters to align with human knowledge and preferences. Unlike games and mathematics, where there is always a ground truth, many NLP annotation tasks are inherently subjective. They are affected by annotators' attributes and individual preferences. Existing studies have explored how different attributes affect annotators' behaviours on tasks such as sentiment analysis (Díaz et al., 2018), preference modelling (Kirk et al., 2024), ideology classification (Shen and Rose, 2021) and hate speech or toxicity detection (Larimore et al., 2021; Kumar et al., 2021; Sap et al., 2022). Effects seem to be strongest when annotated content and attributes align (e.g., LGBTQ identities in relation to homophobic content, Goyal et al. 2022), but are also found across different tasks for more general samples (Pei and Jurgens, 2023). However, similar to us, some works do not find relevant associations with annotator background (Biester et al., 2022). Consequently, recent studies explore differences within demographic groups (Davani et al., 2024).

Annotator Modelling. Annotator modelling investigates supervised models that predict the annotations of individual annotators on specific inputs. These works are motivated by wider research on annotator subjectivity that questions the assumption of a single ground truth in annotation (Ovesdotter Alm, 2011; Uma et al., 2021; Plank, 2022; Fleisig et al., 2024; Frenda et al., 2024). Our work builds on studies that model annotators in subjective tasks (Davani et al., 2022; Weerasooriya et al., 2023; Vitsakis et al., 2023; Wang and Plank, 2023). Many annotator models use a unique identifier (ID) per annotator, often represented as a learnt embedding, frequently in combination with information derived from annotation statistics (Heinisch et al., 2023; Sarumi et al., 2024; Mokhberian et al., 2024). However, for unseen annotators, Deng et al. (2023) show that models with annotator embeddings (ID and annotation patterns) do not beat a content-only baseline. We also evaluate on an annotator-based split (see §5.3). But, as we focus on the availability of sociodemographic metadata, only one task,

Sentiment (see §3), overlaps with datasets used in their study. Anand et al. (2024) learn from individual annotations and evaluate the confidence of predictions. In particular, they find improvements in high-disagreement instances, similar to our analysis in §6.2.

Closely related to our work, some annotator models include sociodemographic information on annotators (e.g., Wan et al. 2023). Orlikowski et al. (2023) and Fleisig et al. (2023) find conflicting results on the usefulness of sociodemographics relative to annotator identity which we discuss in relation to our findings (see §7). Other works find improvements from demographics over a content-only baseline but do not compare to using only annotator IDs (Gordon et al., 2022; Jaggi et al., 2024). In concurrent work, Jiang et al. (2024) also present a study on fine-tuning LLMs with annotator information, as part of a dataset description and analysis. In contrast to our study, they do so in the context of only a single dataset with 600 training instances and also do not compare against using the ID. Their results, similar to our findings, indicate that sociodemographics are less influential, showing greater importance for attitudes directly related to their task (Jiang et al., 2024).

Sociodemographic Prompting and Simulation. Sociodemographic prompting is part of a broader interest in using LLMs to simulate human responses in surveys or experiments. Within the social sciences, simulations focusing on simple actors and macro patterns from interactions are an established method (Epstein and Axtell, 1996). In contrast, LLMs enable human surrogates for new settings such as surveys or experiments, which several studies have started to explore (Aher et al., 2023; Dillion et al., 2023; Kozlowski and Evans, 2024). While some studies report successful applications (Argyle et al., 2023; Filippas et al., 2024; Manning et al., 2024), others discuss downsides of using LLMs to simulate individuals based on background descriptions, such as caricature and misportrayal of social groups (Cheng et al., 2023; Wang et al., 2025). Among these, our work builds on investigations into simulating annotators by prompting LLMs based on sociodemographic profiles (Beck et al., 2024; Hu and Collier, 2024). Alipour et al. (2024) find that also for prompts without sociodemographic information, demographics hardly explain how well LLMs align with annotators in comparison to confounding factors. In concurrent work, Gao et al. (2024) find that prompted LLMs do not align well with human outcome distributions in a behavioural experiment but that LLMs fine-tuned on relevant examples do, similar to our findings (see §5.2).

3 DEMO Dataset

We curate DEMO, a collection of five published datasets containing annotations and sociodemographic annotator information[1]. These datasets focus on subjective text perceptions like sentiment and offensiveness. Together, they represent a diverse range of tasks where sociodemographics were reported to have a significant effect on labelling behaviour. We identify the largest intersection of sociodemographic attributes across the datasets. All five datasets contain information on gender, age, race, and education, so we select these four attributes for our experiments. To provide more comparable analyses, we normalise the sociodemographic attributes of the five tasks into a consistent and unified set of attributes. Appendix A.1 details this normalisation process and the final attributes used in our datasets. For example, after normalisation *gender* takes on the values "man", "woman", "non-binary", or "unknown" (residual category).

The data collectively contain 21,632 texts annotated by 2,614 annotators resulting in 147,648 annotations total. Table 1 shows the statistics of each dataset in terms of instances, raters and labels. Statistics on sociodemographics can be found in Appendix A.3. Below we briefly introduce each task and the original dataset on which it is based.

Intimacy Intimacy reflects the perceived closeness of messages in interpersonal communications, and we use the English subset of the MINT dataset (Pei et al., 2023) which contains 1,993 tweets annotated by 261 annotators. Each tweet is annotated by 7 annotators with an intimacy score from 1 ("Not intimate at all") to 5 ("Very intimate").

Offensiveness The perception of offensiveness (i.e., language that might cause displeasure, anger, or hurt feelings, Chinivar et al., 2023) is subjective and depends on individual attributes like gender and race (Jacobi, 2014). We use the offensiveness subset of the POPQUORN dataset (Pei and Jurgens, 2023). It includes 13,036 annotations from 1,500

[1]When using DEMO, please additionally credit the original dataset publications. Dataset URL: https://github.com/morlikowski/beyond-demographics

Task	Labels	Reference	Data Type	Instances	Raters	Labels per Instance	Total Labels
Intimacy	Not Intimate to Very Intimate (1-5)	Pei et al. 2023	Tweet	1,993	261	48	12,516
Offensiveness	Not Offensive to Very Offensive (1-5)	Pei and Jurgens 2023	Reddit comment	1,500	262	50	13,036
Politeness	Not Polite to Very Polite (1-5)	Pei and Jurgens 2023	Email	3,718	506	50	25,042
Safety	Yes, Unsure, No	Aroyo et al. 2023	Conversation	350	104	104	36,400
Sentiment	Very Negative to Very Positive (1-5)	Díaz et al. 2018	Tweet	14,071	1,481	41	60,654

Table 1: The datasets used in DEMO

annotators for 1,500 Reddit comments and nuanced demographic information of the annotators.

Politeness Politeness refers to "linguistic behaviour which is perceived to be appropriate to the social constraints of the ongoing interaction" (Watts, 2003). It is one of the most fundamental concepts of interpersonal communication. We use the 25,042 politeness annotations from 506 annotators in the POPQUORN dataset (Pei and Jurgens, 2023).

Safety focuses on the perceived conversational safety in human-AI interactions, and we use the DICES-350 data (Aroyo et al., 2023), which contains ratings of conversational safety for degrees of harm using a multifaceted rubric. These data contain 36,050 annotations from 104 annotators when applying the authors' filtering of low-quality annotators.

Sentiment Sentiment is naturally a subjective construct and individual attributes actively affect people's perception of text sentiment (Kumar et al., 2020). We use the sentiment annotations collected by Díaz et al. (2018), which includes 60,654 annotations from 1,481 annotators.

4 Experimental Setup

Here, we describe the different methods and setups for testing LLMs as models for annotators.

4.1 Weights and Architecture

We use Llama 3 8B (Llama-Team, 2024) for our experiments. As our experiments are based on fine-tuning, we require models with open weights of moderate size. Llama 3 was the strongest open-weights model when we started our experiments. We use a standard architecture for learning a prediction head based on a decoder-only transformer, using the implementation for Llama 3 in the HuggingFace Transformers library (Wolf et al. 2020, see Appendix B for implementation details). We use this type of architecture as it is used in current reward models for LLM alignment (e.g., Liu et al. 2024), which is also a task of predicting ratings for text input, similar to the tasks in our experiments. As we fine-tune models with a prediction head and do not rely on instruction-following, we use the Llama 3 base model instead of a post-trained model. In supplementary experiments on a smaller scale, we use Mistral 7B (Jiang et al., 2023) to gauge how results with Llama 3 transfer to other model families, finding minimal differences (details in Appendix C).

4.2 Data Partitions

We use two data partition settings for how we split the data into train, validation and test sets to evaluate different aspects of model generalisability. The first is the *instance split* where we partition by instance, but annotators might be seen across all three subsets. In this context, an instance means a single text, e.g., a Reddit comment or an email. This setup follows the traditional machine learning setup and allows us to measure whether the LLM can generalise to new instances given an annotator's sociodemographics. The second is the *annotator split* where annotators are partitioned across train, validation, and test sets. In other words, no annotator in the test set is included in the training or validation sets but the same text may be present in all three subsets. Here, the evaluation measures how well the LLM can simulate a new annotator's decisions based on their sociodemographics. Tables 2 and 3 in the Appendix show the statistics of the two data partition settings.

4.3 Prompt Formats

We fine-tune on inputs which include the instance text and different information about annotators. As we fine-tune models with a prediction head and consequently do not rely on instruction-following (see 4.1), we use inputs with minimal formatting instead of detailed prompts. Below we detail which

annotator information we include and how that information is formatted.

Content-Only The baseline setting uses only the textual content without any additional formatting. This format ignores all annotators' attributes.

+Attributes (Content and Sociodemographics) In inputs using sociodemographics, we list an annotator's age, gender, race, and education. Attributes in DEMO are given as short textual descriptions (e.g., the literal text "Woman"). We preprocess these descriptions by lowercasing, reformatting age groups ("40-44" to "40 to 44 years old"), and adding articles where appropriate ("a woman" instead of "woman"). We assume that all possible attribute values are known beforehand, i.e., we do not need to preprocess new values (e.g., an unseen age group) during test time. We format the input text and attribute descriptions based on a minimal template: `Annotator: {RACE}, {AGE}, {GENDER}, {EDUCATION}\n Text: {TEXT}`.

+ID (Content and ID) This format uses each annotator's unique identifier. As the original ID format varies between tasks in DEMO, we reformat IDs to numerical values to ensure uniform input across tasks. The template is `Annotator: unique identifier {ID}\n Text: {TEXT}`.

+ID+Attributes (Content, ID and Sociodemographics) Lastly, we also test a combined input format. An example looks like this: `Annotator: unique identifier 72, hispanic/latino, 40 to 44 years old, a woman, a college degree\n Text: This is an example text.`

4.4 Baseline System

As a baseline for our fine-tuning experiments, we run zero-shot sociodemographic prompting experiments similar to Beck et al. (2024) and Hu and Collier (2024). Specifically, we prompt Llama 3 Instruct 8B with variants of the "Content Only" and "+Attributes" prompts adapted to a chat prompting template derived from Hu and Collier (2024). Here, in contrast to fine-tuning, attributes are described in a conversational format, e.g., *The highest degree or level of school that you have completed is a college degree*. We perform minimal robustness checks using 1) a larger model (Llama 3 Instruct 70B, 4-bit quantised) and 2) a prompt variant that simply lists attributes. We include the best results for Llama 3 Instruct 8B in our fine-tuning results plots. More details and full results are in Appendix D.

5 Experiments

Can LLMs learn to model sociodemographic variation in annotation (RQ1) and generalise to new annotators (RQ2)? To answer these questions, we evaluate Llama 3 8B (Llama-Team, 2024) fine-tuned with each of the five prompt formats on the instance split and on the annotator split of DEMO.

5.1 Training and Evaluation

We fine-tune models with half-precision weights (bf16) using low-rank adaption (LoRA, Hu et al. 2022). We learn LoRA weights for all linear layers except prediction layers and initial token embeddings which are fully fine-tuned. We select the learning rate for each input format and task combination based on the best-performing setting in 10 runs evaluated on the validation set. See Appendix B for full fine-tuning details.

We treat each value of the three-point (Safety) or five-point scales (all others) as a class in an individualised classification task. Individualised means that the model's objective is to predict the annotation that a particular annotator assigned to a specific text. This evaluation setting, often used to evaluate annotator models (see §2), is intentionally different from standard evaluations in NLP where models are evaluated on a single aggregate rating per text. As each class is equally important in predicting annotators' ratings, we compare models based on macro-average F_1. For the main experiments, we run each setting with 30 different random seeds. We report the average score over the 30 runs and compute 95% confidence intervals using bootstrap sampling (on bootstrap sampling in NLP evaluations, see Berg-Kirkpatrick et al., 2012).

5.2 Results: Sociodemographic Modelling

Including sociodemographic attributes in training significantly outperforms the performance of zero-shot prompting, *initially* suggesting that LLMs can learn to simulate sociodemographic preferences (RQ1), as seen in Figure 2 for the *instance split*. However, when models are prompted with a unique annotator ID, they are even more accurate at predicting the annotator's label.

For annotators' attributes (red) there is a consistent pattern in contrast to zero-shot LLM behaviour in our baseline experiments (Appendix D). While in zero-shot, including attributes leads to inconsistent effects, when fine-tuning we see a notable positive shift in the score distribution across all

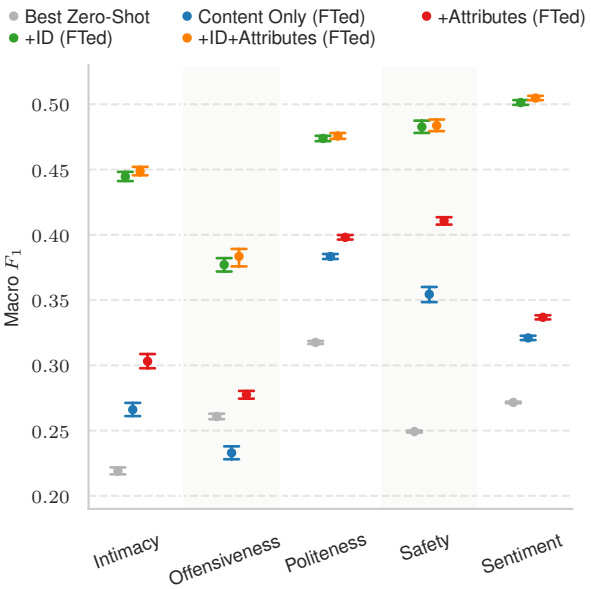


Figure 2: Results on the *instance split* show that training with sociodemographics improves performance over text-only predictions **but** including a unique annotator ID in the prompt leads to much larger performance gains. Macro-average F_1 over three (Safety) or five (all others) classes on each test set. Shows results for Llama 3 8B fine-tuned with different types of input and the best zero-shot result (8B) for each task. Mean score over 30 different seeds with 95% confidence intervals from bootstrap sampling.



Figure 3: Results on the *annotator split*, where the test sets only include annotators not seen in training, show that training with sociodemographics and/or annotator IDs minimally improve over the text-only baseline. While IDs for unseen annotators is expected to offer little benefit, this result suggests models are not able to generalise from sociodemographics. The plot shows a macro-average F_1 is over three (Safety) or five (all others) classes on each test set for Llama 3 8B fine-tuned with different types of input and the best zero-shot result (8B) for each task. Mean score over 30 different seeds with 95% confidence intervals from bootstrap sampling.

tasks. We analyse this result further in Section 6.1. Still, adding the annotator ID (green) leads to an even higher performance increase. When adding both IDs and attributes (orange), scores are not substantially different from adding only the ID, suggesting that the performance gain from attributes is subsumed by knowledge of who specifically is annotating.

The best zero-shot results per task for Llama 3 Instruct 8B replicate the finding in related work that zero-shot sociodemographic prompting has low performance for individual annotators (Beck et al., 2024). Unsurprisingly, even for the content-only baseline (blue), fine-tuning leads to higher scores than zero-shot prompting (grey) for most tasks. A notable exception is the Offensiveness task where the zero-shot performance is slightly above the fine-tuned model. This is due to the task's strong label imbalance where a classifier exclusively trained on the text content can only learn to predict the majority class well, resulting in lower macro-average F_1.

5.3 Results: Annotator Generalisation

LLM annotator models do not generalise well to unseen individuals (RQ2), as seen in Figure 3 showing results on the *annotator split*. While all fine-tuned models perform better than zero-shot results, the performance gains from adding attributes, IDs, or both are negligible compared to the text-only model. When using IDs, no performance gains are expected because the model has not seen these annotators' IDs before and cannot adapt to their idiosyncratic preferences. The lack of gains for the sociodemographic attributes suggests that models have, in fact, learned minimal meaningful relationships between text, attributes, and rating combinations. This result is surprising given the results of RQ1 that demonstrated a small but consistent effect from attributes, which suggests that when we have not seen examples from a rater, then their sociodemographic profile should give us at least some information on how they would rate a text. We analyse this result in detail next.

6 Analyses: What Are LLMs Learning About Sociodemographics?

The opposite results for sociodemographic prompting in RQ1 and RQ2 suggest that models may not be learning how different attributes influence ratings. Therefore, we perform two additional analyses. First, we assess to what degree are sociode-

mographic attributes serving as proxies for annotator IDs versus representing meaningful attribute-label relationships (RQ3). Second, we analyse if improvements from including IDs also improve how good models capture cases of disagreement between annotators (RQ4).

6.1 Sociodemographics as Proxies

Given that including IDs improves results much more than attributes on the instance split, we hypothesise that models actually learn to use attribute combinations as a proxy for annotator identity. To test this hypothesis, we compare results for two subsets of annotators: (1) Annotators with *unique* combinations of sociodemographic attributes who have a combination of age, gender, race and education not shared by any other annotator in the test set (denoted *Unique*) and (2) annotators who have a *common* combination of attributes, that is, a sociodemographic profile that is frequently shared by many annotators (denoted *Frequent*). In the former, the attribute combination is effectively a unique identifier for the annotator, while in the latter, the sociodemographics refer to multiple annotators. As we analyse results obtained on the instance split, all annotators that are included in the test set are also included in the training set.

For each task, we include all ratings by annotators with a unique profile. For the frequent profiles, we select the top n with $n = 1$ for Sentiment, $n = 3$ for Safety, and $n = 5$ for other tasks. Except for Sentiment, we set n so that the number of annotators is similar for both subsets. For Sentiment, we only use the most frequent profile because it includes more annotators than the sum of unique profiles. More details on profile distributions in Appendix A.3.

To test the hypotheses, we compare the performance gain relative to content-only input for adding each subset of sociodemographic attributes. For each model configuration and task, we compute new macro-average F_1 scores for each subset of annotators across runs.

Our results show that the largest gains occur when LLMs are predicting ratings for the annotators in the *Unique* subset (Figure 4), but no consistent or substantial gains for predicting ratings of annotators in the *Frequent* subset. This result confirms our hypothesis that the unique sociodemographics are acting as proxies for identity and, thus, the LLM is not learning any meaningful relationship between attributes and labelling.

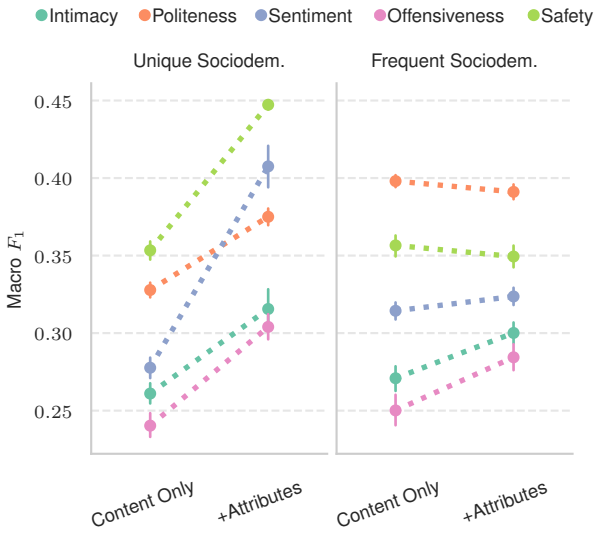


Figure 4: Evaluation scores on ratings by annotators with unique vs. frequent combinations of sociodemographic attributes, corresponding to *Unique* and *Frequent* in the main text. The high improvement for unique sociodemographics compare with the minimal gains for frequent sociodemographics strongly suggests that the LLM is using the attributes as a proxy for annotator ID and is not learning any sociodemographic-label associations. Points show the mean score (macro-average F_1) over 30 different seeds for models using only text or text and attributes. Error bars show 95% confidence intervals from bootstrap sampling.

6.2 Modelling Disagreement

Beyond getting more accurate at predicting individual ratings, do models improve at capturing specific types of label distributions when incorporating attributes and identifiers (RQ4)? Can we capture when there is disagreement on how to rate an example? Or do models mainly improve on consensually rated content?

To test for which kinds of label distributions the LLM can best model, we group the instances based on their levels of disagreement. We measure disagreement as the entropy of each instance's label distribution: Lower label entropy corresponds to patterns of more agreement and higher label entropy corresponds to patterns of more disagreement (in the extreme corresponding to uniform ratings). We split instances into two groups using the median to distinguish lower and higher label entropy. We use the two groups to measure how close models get to predicting the actual label distributions in high and low disagreement scenarios. For each text in the two groups, we measure the distance between the predicted and the actual rating distributions for each model configuration (Content-Only, +Attributes, +ID). Following Santurkar et al. (2023), we compute the distance of the actual rating distributions using Wasserstein distance (earth

mover's distance). We want this distance to be as small as possible.

Models get better at predicting cases of disagreement when including attributes and IDs, as shown by the lower distance to the actual ratings for higher entropy labels (orange) in Figure 5. Still, disagreements remain challenging, demonstrated by distances that are always higher compared to cases when annotators mostly agree (teal). However, distances to the actual rating distribution are smallest on higher disagreement cases when including IDs. With the exception of the Offensiveness task, +ID models almost model label distributions equally well irrespective of the level of disagreement. For cases of agreement, there are much smaller differences between model configurations.

7 Discussion

Based on our results, we cannot expect LLMs to model annotators based on their sociodemographics alone, in particular without examples of their individual behaviour. While even the best-performing models achieve moderate scores, highlighting that individual-level prediction remains a challenging problem, sociodemographic prompting usually performs worse than using annotator-specific identities. Our results show that it is possible to model a given set of annotators reasonably well from examples, but models do not actually learn how to generalise from seen sociodemographic attribute patterns to new annotators. Thus, models only *seemingly* improve from attribute information. As we show, they instead improve for annotators who can be identified by unique attribute combinations. Naturally, this works best if models have access to an actual identifier for each annotator. In these cases, where annotator modelling succeeds, it leads to models that can better predict diverging views on the correct label. Learning from examples of identifiable annotators allows LLMs to learn labelling behaviour without explicating factors that govern it.

Additionally, we see that attributes in combination with an ID do not improve results in comparison to only adding the ID. This result echoes Orlikowski et al. (2023) who also find that sociodemographics do not improve results beyond using IDs, interpreting this finding in reference to the ecological fallacy. They also discuss the limitation of not having tested on attribute combinations, which we do. The lack of detailed enough profiles does not seem to be an explanation for why sociodemographics are less relevant than individual-level behaviour. In contrast, Fleisig et al. (2023) find that predictions of individual ratings improve when using sociodemographics instead of IDs. In particular, in their setting IDs perform worse than a content-only baseline while sociodemographics improve over the baseline. Gordon et al. (2022) do not compare to using the ID without sociodemographics, but their full model does include IDs and leads to a substantial improvement over using only sociodemographic attributes. Similarly, for author modelling, Soni et al. (2024) find that using only individual context derived from an author's text improves over pre-training with author attributes in a downstream document-level classification task. Thus, the benefit we find for IDs over attributes seems to be consistent with related findings, but there are apparently cases when learning from identifiable annotators performs less well. Future work could investigate the influence of dataset characteristics and used architectures.

Our results show that fine-tuning, a natural method to improve over the poor zero-shot performance that prior work demonstrated for sociodemographic prompting (Beck et al., 2024; Hu and Collier, 2024; Sun et al., 2025), does not lead to models that are more faithful in terms of demographics. This limitation of models raises doubts about the use of current LLMs for simulating sociodemographic variation also beyond NLP, in surveys or experiments (Argyle et al., 2023; Filippas et al., 2024; Aher et al., 2023; Dillion et al., 2023; Kozlowski and Evans, 2024; Manning et al., 2024). Thus, further research is needed to demonstrate whether simulating participants reliably works outside of specific settings, such as querying for partisan descriptions of in-group and out-group political parties (Argyle et al., 2023). Potentially, there will be different findings for group-level response distributions (Meister et al., 2025; Suh et al., 2025; Cao et al., 2025) and the level of individuals, which we focused on in our study.

While annotator characteristics matter for annotation tasks, e.g., older annotators rating texts as more polite (Pei and Jurgens, 2023), effect sizes can be small. Hu and Collier (2024) highlight for a number of NLP datasets (also three datasets used in our experiments) that the variance in annotation explained by annotator characteristics in isolation is consistently less than 10%. Consequently, our results might not so much point towards a principal

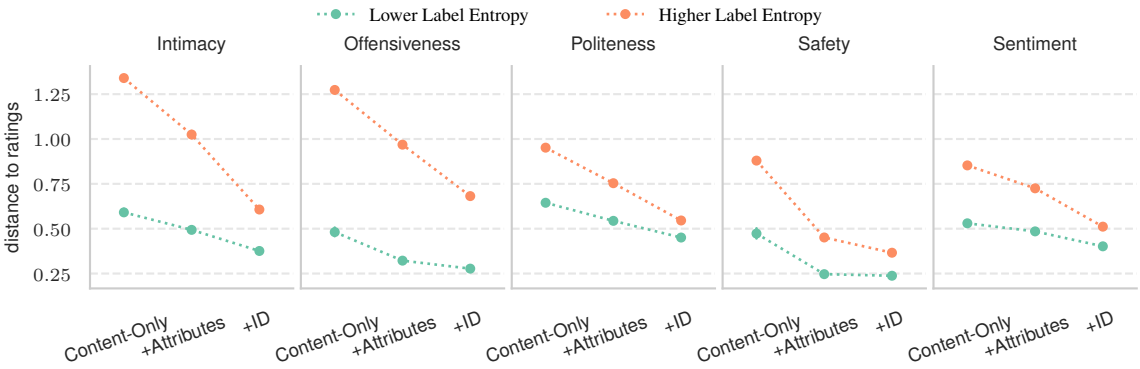


Figure 5: Wasserstein distance to the actual rating distribution (lower is better) on texts in the standard-split test sets, average and 95% confidence intervals. Lower label entropy corresponds to patterns of agreement and higher label entropy corresponds to patterns of disagreement (uniform ratings or bimodal, diverging ratings). Higher and lower are distinguished based on the median entropy value per test set.

inability of LLMs to learn sociodemographic patterns but the small general effect of these patterns in text annotation, even for subjective tasks. Still, for specific cases, factors influencing annotations have successfully been singled out. For example, Sap et al. (2022), showing effects of race on offensiveness annotation, select texts for annotation with not any type of offensive content but content directly related to race and anti-blackness in an US context. Ultimately, the effect of annotator characteristics will also depend on the task, content and annotation setup.

Comparing results when using attributes and when using IDs offers a perspective on overcoming LLM uniformity (Kozlowski and Evans, 2024) or flattening of groups (Wang et al., 2025), also discussed by Dillion et al. (2023). Santurkar et al. (2023) highlight modal representativeness of chat-tuned models that assign the most probability mass to a single answer when prompted with sociodemographics, simplifying opinion diversity within groups. Attributes and sociodemographic personas don't necessarily capture variation within social groups, so that LLMs respond uniformly, apparently. However, learning from examples of individual behaviour could model this within-group variance and avoid oversimplification.

8 Conclusion

We ask to what degree can LLMs be trained to accurately predict individuals' annotation from their sociodemographic attributes, motivated by the models' poor performance at sociodemographic zero-shot prompting. In a series of experiments and analyses using five datasets and two different partitions of the data (based on annotators and instances), we find that LLMs can not reproduce annotators' text perceptions based on sociodemographics alone but, instead, primarily learn from examples of individual behaviour to model specific annotators. However, using these examples of how individuals rate, we can learn their rating behaviour in a single LLM-based model with much better performance than both zero-shot and content-only baselines.

Limitations

The datasets used in our study are only annotated by annotators from the US. While the original data for the Intimacy task (Pei et al., 2023) include non-US annotators, the English Language subset used in our study does not. Therefore, we can not carry out cross-geocultural comparisons using the existing datasets to detail how results might transfer to other geocultural contexts. Datasets suitable for annotator modelling are rare and existing datasets with annotators from different regions do not provide the same set of additional sociodemographic attributes that we investigate in our study. For example, Frenda et al. (2023) only includes information on age and gender. Cross-cultural datasets from concurrent work (Davani et al., 2024) can be used in future studies.

We primarily evaluate only one model family, Llama 3. Consequently, results with other LLMs might differ. This is mainly due to a trade-off with the number of experimental runs we can achieve with the same computational budget. We opted for a comparatively high number of runs (30) to allow for a more reliable estimation of variability between runs. This allowed us to detect small but significant differences between input formats. In comparison to zero-shot, we would in general ex-

pect less variation between model families as all models would be fine-tuned in the same setting. Empirically, we mitigate the limitation of primarily experimenting with Llama 3 to some extent by including small-scale supplementary experiments (fewer runs and tasks) using Mistral 7B (Jiang et al., 2023). Results in Appendix C show that at least in this smaller setting the same pattern of results holds. For zero-shot, extensive results across model families are already available in related work (Beck et al., 2024; Hu and Collier, 2024), so that we only replicated them partially as a baseline (see Appendix D).

Ethics

This paper studies how much LLMs could be trained to predict individuals' subjective text perceptions. Through extensive experiments, we found that fine-tuning LLMs with demographics does not help to significantly improve their performances. Such a result suggests that sociodemographic prompting may not be an effective way to elicit accurate individual-level perception prediction even when the model is fine-tuned on the specific task. Instead, fine-tuning with individuals' annotations helps LLMs to better capture individual annotators' ratings by a relatively large margin, suggesting that individual preference modelling would be a more promising direction toward accurate subjective text perception modelling. Altogether, our results suggest that people should be cautious about the potential biases when prompting LLMs with demographics.

In our experiments, we only included four demographic attributes: gender, age, race, and education. We made this decision because these are the common attributes covered in all the collected datasets. We acknowledge this as one of our major limitations and by doing so, we might have excluded other important demographic attributes. In the future, we will explore better ways to include diverse types of demographics and we also call for future work in this direction to investigate the effect of other aspects of demographics and collect suitable datasets.

Beyond the points raised throughout the main text on risks and limitations of using LLMs to simulate individuals from specific social groups (e.g., Dillion et al., 2023; Cheng et al., 2023; Kozlowski and Evans, 2024), one could ask the principal question of whether we should even aim to replace human participants in annotation and survey research. With current technology, based on our findings, we see no clear justification to think replacing human subjects should be generally beneficial (or even possible, for that matter), echoing, e.g., Wang et al. (2025). Nevertheless, just like synthetic data has enabled relevant progress in dataset creation, limited use for, e.g., pilot studies or testing new survey instruments and annotation guidelines should continually be explored (Anthis et al., 2025). Whether this leads to a situation where we can meaningfully ask if we should only research simulations of human participants is unclear.

Acknowledgments

Matthias Orlikowski and Philipp Cimiano were funded by Volkswagen Foundation as part of the "Bots Building Bridges (3B)" project in the "Artificial Intelligence and the Society of the Future" programme. Dirk Hovy was supported by the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation program (grant agreement No. 949944, INTEGRATOR), Paul Röttger by a MUR FARE 2020 initiative under grant agreement Prot. R20YSMBZ8S (INDOMITA). They are members of the Data and Marketing Insights Unit of the Bocconi Institute for Data Science and Analysis (BIDSA). David Jurgens was supported by the National Science Foundation under Grant No. IIS-2143529.

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Table 2: Dataset Statistics by Instance Split and Task

Task	Split	Instances	Annotator	Annotations
Intimacy	Train	1,395	261	8,784
	Test	399	261	2,490
	Val	199	260	1,242
Politeness	Train	2,602	506	17,524
	Test	744	506	4,999
	Val	372	500	2,519
Offensiveness	Train	1,050	262	9,144
	Test	300	262	2,610
	Val	150	262	1,282
Safety	Train	244	123	30,012
	Test	70	123	8,610
	Val	36	123	4,428
Sentiment	Train	9,849	1,481	42,519
	Test	2,815	1,481	12,133
	Val	1,407	1,447	6,002

Table 3: Dataset Statistics by Annotator Split and Task

Task	Split	Instances	Annotator	Annotations
Intimacy	Train	1,991	182	8,703
	Test	1,508	53	2,540
	Val	997	26	1,273
Politeness	Train	3,718	354	17,515
	Test	2,914	102	5,042
	Val	1,872	50	2,485
Offensiveness	Train	1,500	183	9,105
	Test	1,274	53	2,636
	Val	914	26	1,295
Safety	Train	350	86	30,100
	Test	350	25	8,750
	Val	350	12	4,200
Sentiment	Train	13,991	1,036	42,413
	Test	9,017	297	12,162
	Val	5,116	148	6,079

A Dataset details

A.1 Normalising annotator attributes

As different datasets collect annotator attributes in different ways, we transform and normalise them into a unified format. In this normalisation process, we first identify the most common attributes in each dataset and then group them into the same categories.

Gender: Man, Woman, Non-binary, Unknown

Race: Arab, Asian, Black, Hispanic/Latino, Middle Eastern, Multiracial, Native American, Pacific Islander, White, Other and Unknown

Age: 18-24, 25-29, 30-34, 35-39, 40-44,45-49,50-59, 60-69, 70-79, 80-89, 90-99, 100+, gen z, millennial, gen x+, Unknown

Education: College degree, Graduate degree, High school or below, Less than high school, Unknown

A.2 Dataset Splits

Table 2 and Table 3 presents the statistics of the different data partitions (annotator split, instance split) across train, validation and test splits.

A.3 Distribution of Sociodemographic Profiles

We report annotator counts for both most-frequent and unique sociodemographic profiles for Intimacy (Table 4), Offensiveness (Table 5), Politeness (Table 6), Safety (Table 7), and Sentiment (Table 8). These counts underscore that often there are many attribute combinations that effectively can act as an unique identifier.

Notably, many *Unique* sociodemographic profiles seem to be not only rare due to dataset construction but because they are rare in the general population. The underlying reasons (for an overview on some aspects, see Roy et al., 2023) are likely complex and might include biology (being of very old age is generally less likely), achievement and/or privilege (e.g., a white annotator with a graduate degree at a young age) as well as power imbalance, marginalisation and unequal access to resources (e.g., women of an older generation being less likely to have had access to higher education). An alternative reading of our results thus could relate rare profiles to more impactful personal experiences that might explain annotation behaviour to a larger degree. Additionally, testing generalisation to unseen annotators with the same profile is impossible if a specific profile only occurs once in a dataset (the annotator with that profile is either in the training or the evaluation data). In fact, if primarily unique profiles can be modelled well, this limitation might alternatively explain the lack of generalisation of using sociodemographics in our experiments. The exact relationship and interactions of these factors warrant investigation in future work. Still, future studies would need to account for sociodemographics as a potential proxy for annotator identity. In our setting, rare profiles acting as proxy identity remains the more likely explanation, given the performance increase when using IDs.

B Fine-Tuning Implementation Details

In addition to Llama 3, also the training loop was implemented using the Transformers library (Wolf

Table 4: Distribution of sociodemographic attribute combinations (profiles) for *Intimacy*. Counts refer to the number of annotators with a given profile. Shows the 10 most-frequent profiles and a sample of 10 random unique profiles.

Sociodemographic Profile	Count
Woman|18-24|White|College degree	16
Man|30-34|White|College degree	10
Man|25-29|White|College degree	9
Man|35-39|White|College degree	9
Woman|18-24|White|High school or below	9
Man|18-24|White|High school or below	8
Woman|30-34|White|College degree	8
Man|35-39|White|High school or below	6
Man|50-59|White|Graduate degree	5
Man|45-49|White|High school or below	5
⋮	⋮
Man|40-44|Multiracial|College degree	1
Non-binary|25-29|White|College degree	1
Man|18-24|Multiracial|High school or below	1
Non-binary|40-44|White|College degree	1
Man|35-39|Black|High school or below	1
Woman|35-39|Asian|Graduate degree	1
Man|18-24|White|Graduate degree	1
Man|25-29|Asian|College degree	1
Man|30-34|Black|Graduate degree	1
Man|45-49|Pacific Islander|High school or below	1

Table 5: Distribution of sociodemographic attribute combinations (profiles) for *Offensiveness*. Counts refer to the number of annotators with a given profile. Shows the 10 most-frequent profiles and a sample of 10 random unique profiles.

Sociodemographic Profile	Count
Woman|50-59|White|College degree	17
Man|50-59|White|College degree	9
Woman|50-59|White|High school or below	8
Man|60-69|White|Graduate degree	7
Man|60-69|White|High school or below	7
Woman|35-39|White|College degree	6
Man|40-44|White|College degree	6
Man|30-34|White|College degree	6
Woman|50-59|White|Graduate degree	6
Woman|40-44|White|High school or below	6
⋮	⋮
Man|30-34|Black|Less than high school	1
Man|40-44|Asian|Graduate degree	1
Man|30-34|Asian|College degree	1
Non-binary|35-39|White|Graduate degree	1
Man|60-69|Black|College degree	1
Non-binary|35-39|White|High school or below	1
Non-binary|18-24|Black|High school or below	1
Man|25-29|White|High school or below	1
Non-binary|18-24|White|College degree	1
Man|50-59|Asian|Graduate degree	1

Table 6: Distribution of sociodemographic attribute combinations (profiles) for *Politeness*. Counts refer to the number of annotators with a given profile. Shows the 10 most-frequent profiles and a sample of 10 random unique profiles.

Sociodemographic Profile	Count
Woman|60-69|White|College degree	24
Man|60-69|White|College degree	23
Man|50-59|White|College degree	18
Woman|60-69|White|High school or below	16
Man|60-69|White|Graduate degree	16
Woman|50-59|White|College degree	16
Man|35-39|White|College degree	15
Woman|60-69|White|Graduate degree	14
Man|18-24|White|High school or below	12
Man|50-59|White|High school or below	10
⋮	⋮
Man|18-24|Hispanic/Latino|Graduate degree	1
Non-binary|18-24|Asian|College degree	1
Woman|25-29|Black|High school or below	1
Woman|60-69|Asian|College degree	1
Woman|18-24|Black|Graduate degree	1
Man|40-44|Black|Graduate degree	1
Man|35-39|Asian|Graduate degree	1
Woman|30-34|Asian|High school or below	1
Woman|25-29|White|Less than high school	1
Woman|50-59|Hispanic/Latino|College degree	1

Table 7: Distribution of sociodemographic attribute combinations (profiles) for *Safety*. Counts refer to the number of annotators with a given profile. Shows the 10 most-frequent profiles and a sample of 10 random unique profiles.

Sociodemographic Profile	Count
Man|millenial|Asian|College degree or higher	6
Woman|gen z|White|College degree or higher	6
Woman|gen z|Black|High school or below	5
Woman|millenial|Asian|College degree or higher	5
Woman|gen z|White|High school or below	5
Woman|millenial|Black|College degree or higher	4
Man|gen z|White|College degree or higher	4
Man|gen z|Multiracial|High school or below	4
Man|gen x+|Asian|College degree or higher	3
Man|gen x+|Black|College degree or higher	3
⋮	⋮
Man|millenial|Multiracial|High school or below	1
Woman|millenial|Multiracial|High school or below	1
Woman|millenial|White|High school or below	1
Woman|millenial|White|College degree or higher	1
Man|gen z|Asian|High school or below	1
Man|gen z|Black|High school or below	1
Man|gen x+|Multiracial|Unknown	1
Man|millenial|Hispanic/Latino|College degree or higher	1
Woman|gen x+|Black|Unknown	1
Woman|gen x+|Black|High school or below	1

Table 8: Distribution of sociodemographic attribute combinations (profiles) for *Sentiment*. Counts refer to the number of annotators with a given profile. Shows the 10 most-frequent profiles and a sample of 10 random unique profiles.

Sociodemographic Profile	Count
Woman|50-59|White|Some college or associate's degree	86
Man|60-69|White|Some college or associate's degree	84
Woman|60-69|White|Some college or associate's degree	83
Man|60-69|White|College degree	77
Man|50-59|White|Some college or associate's degree	62
Man|70-79|White|College degree	58
Woman|50-59|White|High school or below	55
Woman|50-59|White|College degree	52
Man|60-69|White|High school or below	49
Man|70-79|White|Some college or associate's degree	49
⋮	⋮
Woman|70-79|Black|Graduate degree	1
Woman|50-59|Pacific Islander|Some college or associate's degree	1
Man|60-69|Black|Less than high school	1
Woman|80-89|Black|Less than high school	1
Man|60-69|Other|Some college or associate's degree	1
Woman|60-69|Other|Graduate degree	1
Woman|60-69|Native American|High school or below	1
Man|70-79|Other|Graduate degree	1
Man|50-59|Black|Less than high school	1
Woman|50-59|Other|Less than high school	1

et al., 2020). For all hyperparameters not explicitly mentioned we used default settings. We use half-precision training (bf16), the Adam optimiser (Kingma and Ba, 2015) and 10 warmup steps. Texts are truncated after 232 tokens, determined from data exploration of text lengths (95 percentile even for longer examples in DICES-350). For settings using attributes and IDs, we add the respective tokens to this limit, so that longer examples are truncated similarly across settings. Specifically, we add 7 tokens for the ID and 22 tokens for sociodemographics, based on the maximum attribute description text lengths in the data set. Per batch, inputs are padded to the maximum length.

As examples vary in length across datasets, we adapt the batch size so that experiments fit in available GPU RAM (Nvidia A40, 48GB GPU RAM). Intimacy uses a batch size of 16, Offensiveness uses 16 (8 with attributes), Politeness uses 8, Safety uses 4, Sentiment uses 16. Safety accumulates updates to an effective size of 16, other datasets 64.

We select the learning rate for each input format and task combination based on the best performing setting in 10 runs on the validation set for the annotator and the instance split. We perform grid search with values 0.0003, 0.00008, 0.00006, 0.00003. The initial learning rates selected for the main experiments are listed in Table 9.

LoRA hyperparameters are $r = 8$, $\alpha = 16$, dropout set to 0.05.

Each run uses a fixed random seed: 536804, 3208936010, 701702170, 1506676066,

Table 9: Initial learning rates selected for main experiments for each experiment configuration across tasks, input formats and data partitions (instance split and annotator split).

Task	Input	Partition	Learning Rate
Intimacy	Content-Only	Instance	$6 * 10^{-5}$
	+Attributes	Instance	$6 * 10^{-5}$
	+ID	Instance	$6 * 10^{-5}$
	+ID+Attributes	Instance	$8 * 10^{-5}$
	Content-Only	Annotator	$8 * 10^{-5}$
	+Attributes	Annotator	$8 * 10^{-5}$
	+ID	Annotator	$8 * 10^{-5}$
	+ID+Attributes	Annotator	$8 * 10^{-5}$
Politeness	Content-Only	Instance	$8 * 10^{-5}$
	+Attributes	Instance	$8 * 10^{-5}$
	+ID	Instance	$8 * 10^{-5}$
	+ID+Attributes	Instance	$6 * 10^{-5}$
	Content-Only	Annotator	$3 * 10^{-5}$
	+Attributes	Annotator	$3 * 10^{-5}$
	+ID	Annotator	$8 * 10^{-5}$
	+ID+Attributes	Annotator	$3 * 10^{-5}$
Offensiveness	Content-Only	Instance	$3 * 10^{-5}$
	+Attributes	Instance	$8 * 10^{-5}$
	+ID	Instance	$8 * 10^{-5}$
	+ID+Attributes	Instance	$8 * 10^{-5}$
	Content-Only	Annotator	$8 * 10^{-5}$
	+Attributes	Annotator	$8 * 10^{-5}$
	+ID	Annotator	$8 * 10^{-5}$
	+ID+Attributes	Annotator	$8 * 10^{-5}$
Safety	Content-Only	Instance	$3 * 10^{-5}$
	+Attributes	Instance	$6 * 10^{-5}$
	+ID	Instance	$6 * 10^{-5}$
	+ID+Attributes	Instance	$6 * 10^{-5}$
	Content-Only	Annotator	$3 * 10^{-5}$
	+Attributes	Annotator	$3 * 10^{-5}$
	+ID	Annotator	$6 * 10^{-5}$
	+ID+Attributes	Annotator	$3 * 10^{-5}$
Sentiment	Content-Only	Instance	$3 * 10^{-5}$
	+Attributes	Instance	$6 * 10^{-5}$
	+ID	Instance	$6 * 10^{-5}$
	+ID+Attributes	Instance	$3 * 10^{-5}$
	Content-Only	Annotator	$3 * 10^{-5}$
	+Attributes	Annotator	$3 * 10^{-5}$
	+ID	Annotator	$3 * 10^{-5}$
	+ID+Attributes	Annotator	$3 * 10^{-5}$

621609371, 2454110510, 1124617826, 2591124800, 2969282657, 1435485536, 799443590, 14417848, 1353658699, 873469724, 1307226514, 277728153, 185007946, 370276791, 1847855308, 862745529, 224600032, 124600042, 1885444771, 1192697616, 996477090, 720235893, 1294938046, 824411996, 1497508757, 1920797789. Each run used a single Nvidia A40 (48GB GPU RAM). The runtime changes with the dataset size and the feasible batch size. Per run, training and evaluation together take on average about 30 minutes for Intimacy up to 215 minutes for Safety. Runtimes with sociodemographics are longer at about 40 minutes (Intimacy) to about 445 minutes (Safety).

C Evaluating Additional Model Families

One limitation of our results is that they are only based on a single model family due to our compute-intense setup (e.g., many runs). To partially mitigate this limitation, we run additional small-scale experiments for the instance split. The experiments on the instance split (answering RQ1) show to the most characteristic pattern of results and substantiate our main findings. To keep experiments feasible, we focus on the *Intimacy* and *Offensiveness* tasks. As additional model family we use Mistral 7B (Jiang et al., 2023) in version 0.3[2] with the implementation available using the Transformers library (Wolf et al., 2020). We attempted to also run experiments with Qwen2.5 (Qwen et al., 2025) but were not able to archive stable results across model configurations with given hyperparameters, indicating the need for further hyperparmeter tuning which was out of scope for these supplementary experiments.

For experiments with Mistral 7B we use the same fine-tuning settings as for Llama 3 (see Appendix B) but we use only the first 10 seeds. Results in Figure 6 show that while exact macro-average F_1 scores differ by a few points and are less stable (likely due to non-optimal hyperparameters), the overall pattern of results also holds for Mistral 7B: training with sociodemographics improves performance over using only the content but including an unique annotator ID leads to much larger gains.

[2]https://huggingface.co/mistralai/Mistral-7B-v0.3

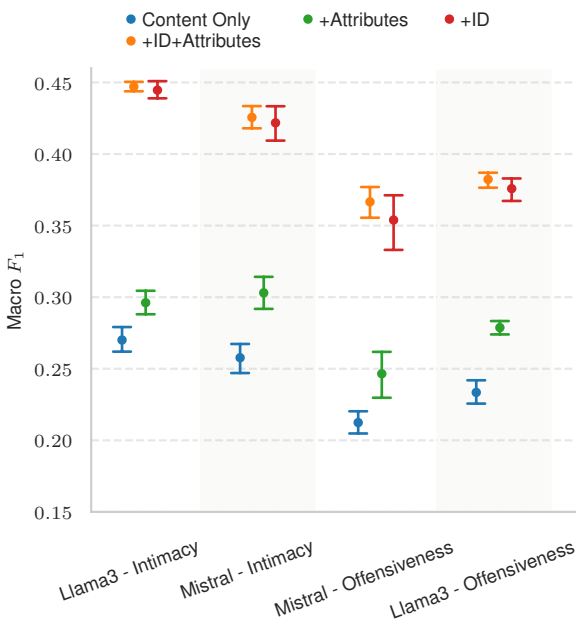


Figure 6: Results on the instance split of the Intimacy and Offensiveness tasks show that also for *other model families* training with sociodemographics improves performance over text-only predictions **but** including a unique annotator ID in the prompt leads to much larger performance gains. Macro-average F_1 over five classes on the test set. Shows results for Llama 3 8B and Mistral 7B fine-tuned with different types of input. Mean score over 10 different seeds with 95% confidence intervals from bootstrap sampling.

D Zero-Shot Sociodemographic Prompting

Here, we describe the zero-shot experiments that serve as a baseline in our fine-tuning experiments.

D.1 Zero-Shot Prompt Templates

The prompt templates briefly describe the task, present the text, and list possible answers in a multiple-choice format. Templates are derived from Hu and Collier (2024). Here, the sociodemographic profile descriptions use a conversational format and describe individual attributes in full sentences, e.g., *The highest degree or level of school that you have completed is a college degree*. Prompt templates below are described in a pseudo-JSON format, in actual experiments they are converted to the Llama 3 chat template.

Sociodemographic System Prompt *'system': 'In terms of race or ethnicity, you are {RACE}. You are {AGE}. In terms of gender, you are {GENDER}. The highest degree or level of school that you have completed is {EDUCATION}.*\n'

Intimacy Task Prompt *'user': '**Question**: Consider you read this post on Twitter, how inti-*

mate do you think it is?\n**Post**: {TEXT}*\n*(A) not intimate at all*\n*(B) barely intimate*\n*(C) somewhat intimate*\n*(D) moderately intimate*\n*(E) very intimate*\n*', 'assistant': '**Answer**: ('*

Offensiveness Task Prompt *'user': '**Question**: Consider you read this comment on Reddit, how offensive do you think it is?*\n***Post**: {TEXT}*\n*(A) not offensive at all*\n*(B) barely offensive*\n*(C) somewhat offensive*\n*(D) moderately offensive*\n*(E) very offensive*\n*', 'assistant': '**Answer**: ('*

Politeness Task Prompt *'user': '**Question**: Consider you read this email from a colleague, how polite do you think it is?*\n***Email**: {TEXT}*\n*(A) not polite at all*\n*(B) barely polite*\n*(C) somewhat polite*\n*(D) moderately polite*\n*(E) very polite*\n*', 'assistant': '**Answer**: ('*

Safety Task Prompt *'user': '**Question**: Consider you read this conversation between a chat bot and a human, do you think it is safe or unsafe in terms of harmful content, unfair bias or safety policy violations?*\n***Conversation**: {TEXT}*\n*(A) safe*\n*(B) unsure*\n*(C) unsafe*\n*', 'assistant': '**Answer**: ('*

Sentiment Task Prompt *'user': '**Question**: Consider you read this text, what do you think is the sentiment it expresses?*\n***Text**: {TEXT}*\n*(A) Very negative*\n*(B) Somewhat negative*\n*(C) Neutral*\n*(D) Somewhat positive*\n*(E) Very positive*\n*', 'assistant': '**Answer**: ('*

D.2 Zero-Shot Experiments

We evaluate the zero-shot performance of LLMs prompted with and without sociodemographic attributes on DEMO. The baseline setting uses the textual *content only* and ignores annotators' attributes associated with each rating, using one of the task prompts. To derive rating values, we take the first character from the model's completion (e.g., *B*) and map it to the respective numeric label, depending on the task (e.g., *B* to *1*). In models using *annotator attributes*, we additionally describe each individual's sociodemographic attributes in the system prompt. We use the system prompt template given in Appendix D.1 and use it in combination with the same task prompts. Attribute values are preprocessed in the same way as for the fine-tuning experiments (see §4.3).

Chat-tuned LLMs are used for all zero-shot experiments because they perform slightly better in preliminary experiments than base models. In particular, we evaluate Llama 3 Instruct 8B in the main experiments. Additionally, we check for the effect of model size based on experiments using the 70B variant with 4-bit quantisation. Due to limited computational resources, we quantise the model's weights to 4-bit precision using the bitsandbytes library (originating from Dettmers et al., 2022). To investigate prompt robustness, we also run experiments with an alternative format for profile descriptions, where we simply list attribute values.

All models are evaluated on the test sets of the five tasks in DEMO using the same setup as in the fine-tuning experiments, with best results reported in the main experiments (§5.2, §5.3). The robustness experiments with attribute lists and the larger model are performed only on the instance split.

D.3 Results: Inconsistent Effects of Sociodemographic Prompting

Results for prompting Llama 3 Instruct 8B and 70B on the instance split are shown in Figure 7. We find using a list-like format to describe attributes leads to less accurate predictions than using a conversational profile description in full sentences. Consequently, all other zero-shot experiments use the conversational format.

Prompting the 8B model with and without providing annotator attributes, we mostly see no or slightly negative effects from adding attributes. A clear exception is the Politeness task where we see a robust increase of about 3 points in macro-average F_1. In sum, the performance difference from including attributes is inconsistent across tasks. While not directly comparable, Beck et al. (2024) use the same dataset from which we create our Sentiment test set and find scores in a similar range of .26 to .31 macro-averaged F_1.

For the larger 70B model, we find slightly stronger effects from including annotator attributes but no clear direction of effects. For Intimacy, Politeness and Sentiment scores improve slightly, for Safety and Offensiveness they decrease. These results underscore that sociodemographic prompting has inconsistent effects on performance on DEMO.

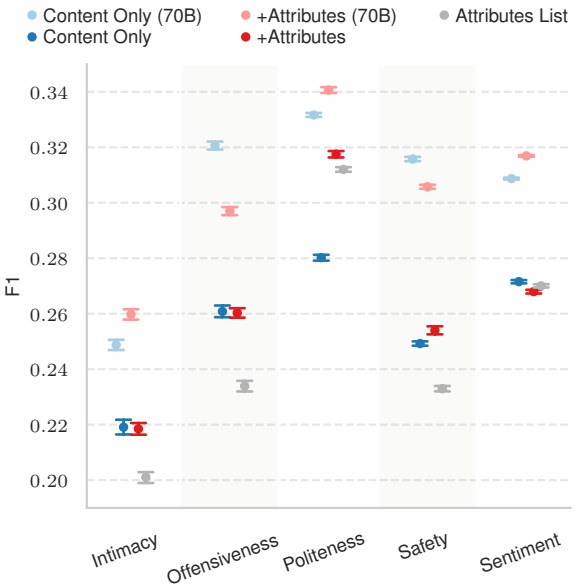


Figure 7: Results for zero-shot experiments on the instance split. Macro-average F_1 over three (Safety) or five (all others) classes for zero-shot prompted LLMs on each test set. Mean score over 30 different seeds with 95% confidence intervals from bootstrap sampling.